

X	The input to a neural network or an exogenous variable in a causal graph. A specific value of X is denoted x .
Y	The output of a neural network or outcome node in a causal graph. A specific value of Y is denoted y .
$\mathcal{C}$	The computation graph of a neural network. Also used to refer to the neural network itself.
Z	A generic placeholder referring to any possible representation in a neural network between X and Y .
ℓ	A layer of $\mathcal{C}$.
$\mathbf{h}^\ell$	The representation vector at the output of layer ℓ .
$\mathbf{h}^{\ell\text{-MLP}}, \mathbf{h}^{\ell\text{-Attn}}$	The vector output of the MLP or attention block, respectively, at layer ℓ .
$\mathbf{h}_i^\ell$	A neuron in $\mathbf{h}^\ell$.
h_i^ℓ	A scalar activation of neuron $\mathbf{h}_i^\ell$.
A_i^ℓ	An attention head in layer ℓ .
$\mathbf{a}_i^\ell$	The vector output (attention score, equivalent to $Q \cdot K$ before the softmax) of A_i^ℓ .
d	The size of $\mathbf{h}^\ell$.
$\mathcal{H}$	A causal graph.
V	The set of nodes in causal graph $\mathcal{H}$ between X and Y .
V_i	A node in V . A specific value of V_i is denoted as v_i .
E	The set of edges in causal graph $\mathcal{H}$.
$E_{i,j}$	An edge in E drawn from V_i to V_j .

Table 1

Table of notation grouped by whether the terms refer to concepts in neural networks, causal graphs, or both.

or event; in neural networks, it can correspond to any component (or combination thereof), as described below and in §5. An **edge** $E_{i,j} \in E$ encodes a causal relationship between nodes, where the source is the **cause** and destination is the **effect**.³ For example, if edge $E_{i,j}$ is drawn from one neuron V_i to another V_j , this indicates that V_i has significant counterfactual influence over V_j . Note that in a causal chain (a connected path in a causal graph), any node V_j can simultaneously function as both a cause of some downstream node V_k and an effect of a prior node V_i .

The abstraction of causal graphs extends naturally to neural networks: the computation graph of a model $\mathcal{C}$ is, by definition, the full causal graph that explains how inputs X (an exogenous variable in the causal graph) are transformed into a probability distribution over outputs Y (an outcome or leaf node in the causal graph). A causal node can correspond to any unit or intermediate representation Z produced by the network—for example, a neuron, a full layer, an attention head, or even some grouping of these. An edge encodes a causal relationship between any two nodes in the network, where the only restriction is that the source of the edge come before the destination in the computation graph. We summarize the notation we use for describing (components of) causal graphs and computation graphs in Table 1.

Each node can be viewed as a causal mediator that has some functional role in explaining how X is transformed into Y . The main challenge of mechanistic interpretability studies, then, is to define a mapping from components Z in computation

³ In interpretability studies, it is also frequently required that edge E_i have strong influence on the final target behavior or output Y , rather than just the downstream intermediate component V_j .

graph $\mathcal{C}$ to a high-level causal graph $\mathcal{H}$ consisting of nodes and edges (V, E) that explains how the model performs some specific behavior. This entails deciding which of the components $\{Z_i\}_{i=1}^N$ should be filtered out from the nodes of the causal graph V , and optionally filtering out edges between these nodes.⁴ This survey focuses on how the type of component $Z \in \mathcal{C}$ will affect one’s findings; this is discussed in detail in §5.

In the causality literature, a **mechanism** is defined as a causal chain from cause X to effect Y (Salmon 1984; Pearl 2009). The mechanistic interpretability literature, while closely related to causal interpretability,⁵ does not enforce this causally grounded definition of mechanism (cf. Miller, Chughtai, and Saunders 2024; Nanda, Lee, and Wattenberg 2023). The overlap between mechanistic and causal interpretability is significant, but not total: for example, sparse autoencoders (Bricken et al. 2023; Cunningham et al. 2024) are correlational, but are common in mechanistic interpretability, as they can reveal the concepts encoded in a model component without requiring the researcher to hypothesize what these concepts are ahead of time. Meanwhile, methods like LIME (Ribeiro, Singh, and Guestrin 2016b) involve interventions to input variables, but not the internals of a model. We believe that the causal definition of “mechanism” is an actionable one that makes the main challenge of mechanistic interpretability more precise—to reverse-engineer algorithmic understanding or control of model behaviors, where “algorithm” is equivalent to a task-specific causal graph $\mathcal{H}$ explaining how the model $\mathcal{C}$ performs a given task.

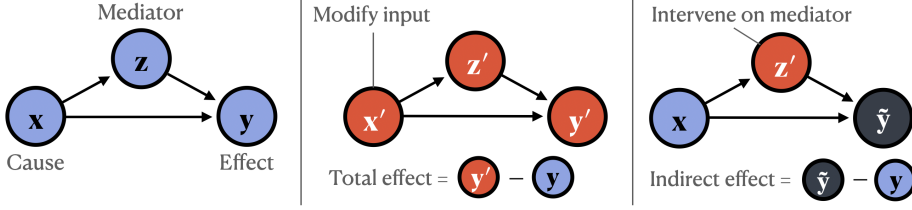
Counterfactual interventions. In interpretability, “causal method” generally refers to a method that employs **counterfactual interventions** (Lewis 1973) to some part of the model or its inputs. Early interpretability work focused on interpreting model decision boundaries by intervening on the inputs X to the network (e.g., Ribeiro, Singh, and Guestrin 2016b), but contemporary work is primarily concerned with understanding which intermediary model components Z are responsible for some behavior Y —i.e., finding the model components $Z \in \mathcal{C}$ from the low-level computation graph to keep as nodes $V \in \mathcal{H}$ in the high-level causal graph (e.g., Geiger et al. 2021; Hanna, Liu, and Variengien 2023a).

Causal mediation analysis (Pearl 2001) provides a framework for performing counterfactual interventions and interpreting their results. Given input X , output Y , and a causal graph consisting of many intermediate nodes V between X and Y , the causal influence of an intermediate node (mediator) $V_i \in V$ on the output Y is quantified as V_i ’s **indirect effect** (IE; Pearl 2001; Robins and Greenland 1992). This metric is based on the notion of counterfactual dependence, where one measures the difference in some target metric m before and after intervening on the mediator. In practice, m is typically (but not always) a function of the model’s output Y .⁶ More precisely, one starts by measuring m

⁴ For some studies, it is sufficient to discover unordered sets of causally relevant components. In such cases, we assume that the causal graph is fully connected (where it is possible given the computation graph’s directionality).

⁵ To be more precise, we will classify any study that aims to understand a model via understanding the roles of its components or its inner representations as mechanistic interpretability. We will classify any study that employs counterfactual interventions to a model’s inputs and/or representations (resulting in states the model would not naturally have taken) as causal interpretability.

⁶ m can be any scalar value, including the value of an intermediate causal variable $\in V$, or even the aggregated values of multiple intermediate variables. For example, ACDC (Conmy et al. 2023) recursively computes a circuit by first finding components with high IE on Y (where m is derived from Y), and then finding components with high IE on those components (where m is now derived from intermediate components Z), and so on.

**Figure 2**

Visual summary of causal mediation analysis. Given input $X = x$ and the resulting output (model prediction) $Y = y$, and another input $X = x'$ that results in different output $Y = y'$, we can compute the total effect of changing x to x' as $y' - y$. In neural networks, there exist components Z that mediate the influence of X on Y . A common way to quantify the importance of Z is by measuring its **indirect effect** (Eq. 1), where, given $X = x$, one sets Z to some counterfactual value z' . In this figure, we set Z to what it would have been given x' ; this results in $Y = \tilde{y}$. One can then measure the indirect effect as $\tilde{y} - y$.

given a normal run of the model on input $X = x$, where V_i takes its natural value(s) v_i . We then compare this to m given $X = x$ where we intervene to set mediator V_i to some alternate (counterfactual) value v'_i :⁷

$$\text{IE}(m; X, x; V_i, v_i, v'_i) = m(V_i = v_i \mid X = x) - m(\text{do}(V_i = v'_i) \mid X = x), \quad (1)$$

where “do” is an operation where the experimenter intervenes, setting a value in the experiment to one that it would not naturally have taken without the experimenter’s involvement. See Figure 2 for an illustration of the indirect effect given a neural network component Z .⁸

From intervention to evaluation. The tools of causal mediation analysis allow us to quantify the causal influence of a mediator on model behavior. However, identifying influential components is not sufficient for mechanistic understanding on its own. Depending on the goal of a study, we may define additional criteria by which the appropriateness or quality of a mediator may be judged. In the following subsection, we outline an evaluation framework that will help frame the following survey of mediator types used in existing research.

2.1 Evaluation Criteria

The goals of a mechanistic interpretability study determine how success should be defined. We identify three primary goals: (1) explaining model behavior, (2) verifying a mechanistic hypothesis, and (3) localization and editing. Below, we define criteria by which we can compare the utility of mediator types.

⁷ Appendix A surveys methods for sourcing v'_i . This value can come from alternate inputs where the answer is flipped, means over many inputs, or arbitrary constants (typically 0).

⁸ The choice of counterfactual v'_i significantly affects which components will be identified as influential. Common approaches for selecting v'_i include using values from alternative inputs x' , applying a constant (e.g., 0), or adding noise ϵ , where ϵ is typically scaled according to the norm or variance of v_i . These are conceptually distinct operations that will uncover different kinds of components; we provide a more detailed discussion in Appendix A. We leave this to the appendix, as this survey is concerned with how results are affected by the choice of mediator Z —a topic distinct from the choice of intervention value v'_i .